

GalaxEye Technical Assignment

Satellite AI Research Intern

SAR-to-EO Image Translation Using Progressive Generative Models

Technical Report

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July 2026

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1 Abstract

Synthetic aperture radar (SAR) and electro-optical (EO) imagery observe the same Earth surface through different physical sensing processes. SAR provides all-weather and day-night imaging, but it is affected by speckle, viewing geometry, layover, and shadow. Optical imagery provides intuitive colour, texture, and semantic cues, but it is affected by cloud cover, illumination, and acquisition gaps. This project studies SAR-to-EO image translation as a conditional generative modelling task on paired Sentinel-1/Sentinel-2 image patches.

The work is structured as a progressive experimental study rather than a single architecture trial. Five models are trained and compared: a deterministic U-Net with L1 loss, Pix2Pix, Pix2Pix with perceptual and feature-matching losses, an attention ResUNet GAN, and a final fusion GAN with hinge adversarial loss, gradient consistency, and exponential moving average checkpointing. The dataset contains 16,000 paired patches across four terrain classes. The split is scene-disjoint, with 10,775 training patches, 2,514 validation patches, and 2,711 test patches.

Final evaluation was performed on 200-sample validation and test subsets with PSNR, SSIM, MAE, LPIPS, FID, and runtime. On the test subset, no single model dominates all metrics. M4 obtains the best PSNR (13.994), M1 obtains the best SSIM (0.211), M5 obtains the best MAE (0.164), M2 obtains the best LPIPS (0.615), and M3 obtains the best FID (261.647). This disagreement is the central finding of the project: pixel fidelity, perceptual similarity, and distributional realism reward different model behaviours. A controlled ablation of the final model further shows that perceptual supervision is the most important tested component under the reduced training budget, while attention is not automatically beneficial in short runs.

The final conclusion is therefore not that SAR-to-EO translation is solved by one architecture. Instead, the project builds a defensible experimental ladder, identifies the trade-offs introduced by each modelling decision, and shows why future work should move toward calibrated SAR inputs, structure-aware supervision, and uncertainty-aware conditional diffusion.

2 Introduction

Remote sensing imagery is a fundamental source of Earth-observation information and supports environmental monitoring, agricultural assessment, urban planning, disaster response, land-cover analysis, and infrastructure monitoring. Among remote sensing modalities, SAR and optical imagery are especially complementary. SAR provides all-weather and day-night observation, while optical imagery provides colour, texture, and semantic information that is easier to interpret visually.

The two modalities also have different limitations. Optical images are affected by cloud cover, illumination, atmospheric conditions, and acquisition timing. SAR is robust to weather and lighting, but contains speckle noise and geometric effects such as layover and shadow. Because of platform constraints, weather interference, and scheduling differences, paired SAR and optical observations are not always available. SAR-to-EO image translation attempts to reduce this gap by synthesizing an optical-like image from a SAR input.

Generative models provide a natural framework for this task. Conditional GANs, Pix2Pix-style paired translation, CycleGAN-style unpaired translation, hybrid GANs, and diffusion models have all been used for image-to-image translation. These methods aim to preserve source structure while synthesizing target modality appearance. However, SAR-to-EO translation is underdetermined: SAR backscatter does not uniquely determine optical colour, illumination, material appearance, or fine texture. A generated EO image should therefore be treated as a plausible prediction, not as a physically observed optical measurement.

This report studies SAR-to-EO translation through a progressive experimental framework. Instead of choosing one model at the beginning, we train and evaluate five models on the same scene-disjoint dataset. The progression starts with a deterministic U-Net baseline, moves to Pix2Pix, adds perceptual and feature-matching losses, introduces attention and residual generation, and ends with a refined final GAN using gradient consistency and EMA checkpointing. This design makes the project more than a training exercise: each stage tests a modelling hypothesis and exposes a different failure mode.

The objective is to analyse the SAR/optical data, justify architecture choices using the literature, compare multiple generative models, and explain both successful and failed behaviours. The report is organized as follows. Section 3 reviews the relevant literature. Section 4 presents the dataset and EDA. Section 5 describes the experimental design. Sections 6–8 report training, validation/test evaluation, and controlled ablation. Sections 9–10 discuss qualitative results and failure analysis. Sections 11–13 present limitations, future work, and conclusions.

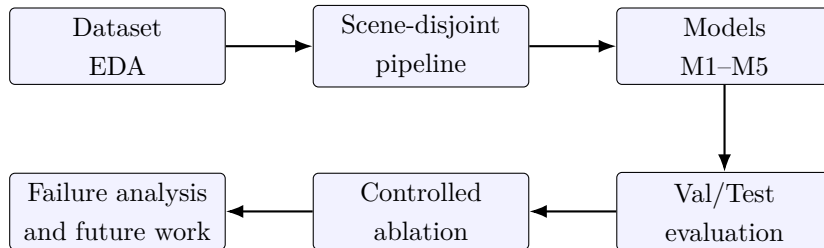


Figure 1: Project workflow. The report follows a complete experimental loop: inspect the data, build a reproducible pipeline, train progressive models, evaluate on validation and test splits, run a controlled ablation, and analyse failures.

3 Literature Survey

3.1 SAR–Optical Translation as a Remote Sensing Problem

SAR–optical image translation is motivated by the complementary nature of radar and optical sensing. SAR provides all-weather and day-night observation, while optical imagery provides colour, texture, and semantic information that is easier to interpret. The SEN1-2 dataset demonstrated the usefulness of paired Sentinel-1/Sentinel-2 imagery for cross-modal learning, including SAR–optical matching and artificial optical image generation [1].

Unlike ordinary image restoration, SAR-to-EO translation is a cross-modal synthesis problem. SAR backscatter and optical reflectance are produced by different physical mechanisms, so the mapping is not one-to-one. Early SAR-to-optical work by Fu et al. showed that adversarial learning can support assisted SAR interpretation and reciprocal SAR–optical generation [2, 3]. These works motivate our use of paired generative models, but also warn that outputs should be evaluated as plausible translations rather than ground-truth observations.

3.2 GANs, Conditional GANs, and Pix2Pix

Generative adversarial networks were introduced by Goodfellow et al. as a two-player learning framework between a generator and a discriminator [4]. Conditional GANs extend this idea by conditioning generation on an input signal [5]. This makes them suitable for image-to-image translation, where the output must correspond to a specific input.

Pix2Pix by Isola et al. is the natural paired translation baseline for this project [6]. It combines a U-Net generator, a PatchGAN discriminator, and an L1 reconstruction loss. The U-Net architecture preserves spatial layout through encoder–decoder skip connections [7], while PatchGAN encourages local realism. In this project, M1 isolates the U-Net reconstruction baseline, while M2 adds the adversarial Pix2Pix objective.

3.3 Feature Matching and Perceptual Losses

Pixel losses often produce blurry outputs when the target has multiple plausible appearances. Pix2PixHD introduced feature matching to stabilize adversarial training and improve high-resolution synthesis [8]. Perceptual losses compare feature activations rather than only pixel values, and LPIPS showed that learned feature distances correlate better with perceptual similarity than simple pixel metrics [9]. These ideas directly motivate M3, which adds perceptual and feature-matching terms to Pix2Pix.

3.4 Residual Learning, Attention, and Stable Discriminators

Remote sensing images contain structures such as roads, fields, rivers, ridges, and urban blocks. Preserving these structures is essential for useful SAR-to-EO translation. Residual learning improves optimization in deep networks [10], and attention mechanisms such as CBAM refine spatial and channel features [11]. Spectral normalization stabilizes discriminator training by controlling layer norms [12]. M4 uses these ideas through an attention ResUNet generator and spectral-normalized PatchGAN discriminator.

3.5 CycleGAN and Why It Was Not the Final Direction

CycleGAN supports image translation when paired data are unavailable by using adversarial and cycle-consistency losses [13]. It is important for remote sensing because perfectly paired SAR–optical data are not always available. In this assignment, however, paired Sentinel-1/Sentinel-2 patches are available. Using an unpaired method as the main final model would discard useful supervision. Therefore CycleGAN is discussed as literature context, but the main experimental path focuses on paired translation.

3.6 Diffusion Models as the Next Step

Diffusion models generate images through iterative denoising [14]. Latent diffusion improves scalability by operating in a compressed latent space [15]. Recent SAR-to-optical diffusion work has explored colour-supervised diffusion and accelerated adversarial consistency distillation for SAR-to-optical translation [16, 17]. These methods are attractive because SAR-to-EO translation is naturally one-to-many: one SAR input may correspond to multiple plausible optical appearances. In this project, diffusion is positioned as future work because it requires more compute, slower sampling, and careful uncertainty evaluation.

3.7 Position of This Work

This project builds on the literature by turning architecture selection into an experimental ladder. We start with U-Net reconstruction, then add adversarial learning, perceptual and feature-matching losses, attention and residual generation, and final stabilization with gradient consistency and EMA. This structure lets us defend the final model and, more importantly, explain why each modelling step succeeds or fails.

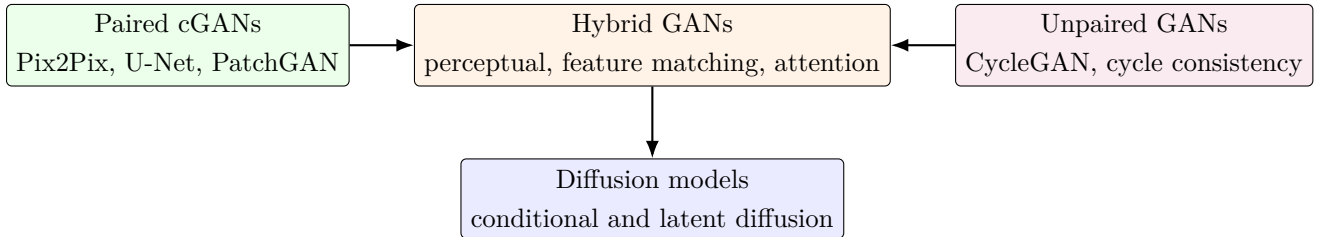


Figure 2: Project-specific taxonomy of generative models relevant to SAR-to-EO translation. The schematic summarizes original method families rather than reproducing copyrighted survey figures.

4 Dataset and Exploratory Data Analysis

4.1 Dataset Overview

The dataset is Sentinel-1/2 SAR and Optical Image Pairs Segregated by Terrain. Each sample contains a SAR patch and paired optical RGB patch. The EDA identifies 16,000 paired images, 32 unique scenes, and four terrain classes: agriculture, urban, barren land, and grassland. The dataset is balanced with 4,000 patches per terrain. All images are 256×256 pixels.

Table 1: Dataset summary from Script 01.

Property	Value
Total paired patches	16,000
Unique scenes	32
Train patches	10,775
Validation patches	2,514
Test patches	2,711
Terrain classes	4
Resolution	256×256
Corrupt images	0
Missing EO pairs	0
Exact duplicate groups	2
Scene-disjoint split passed	True

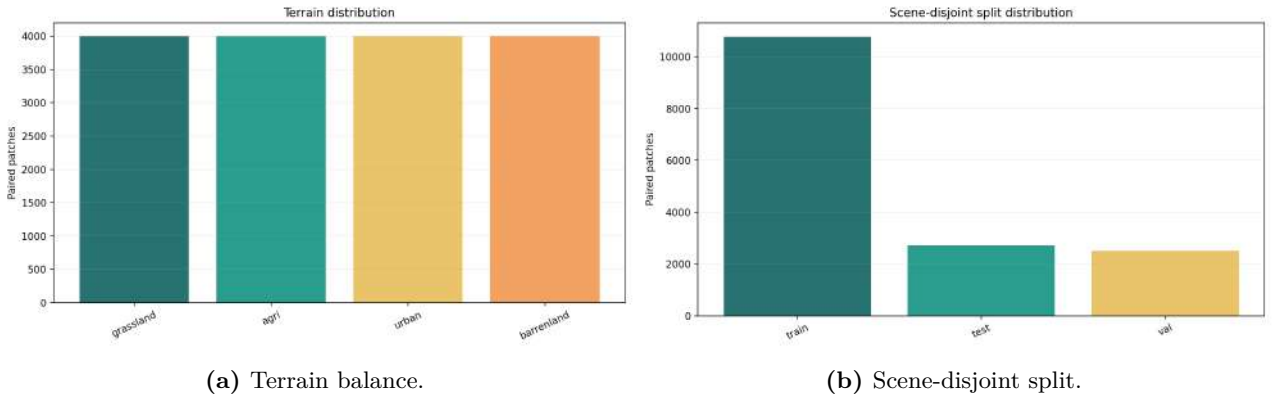
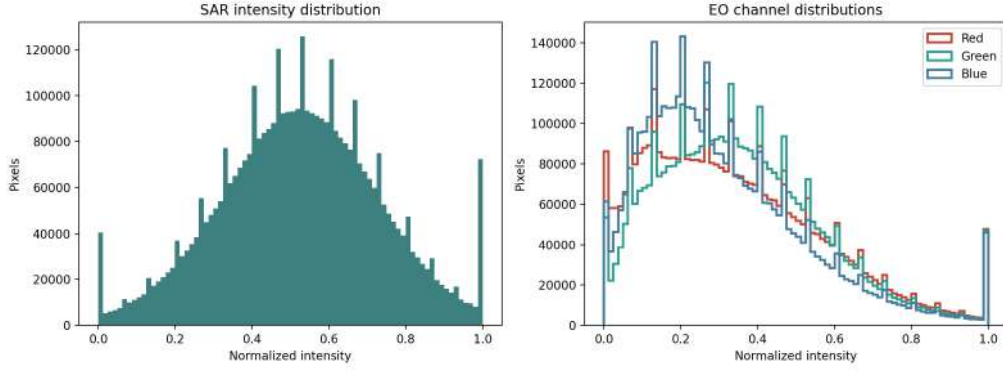


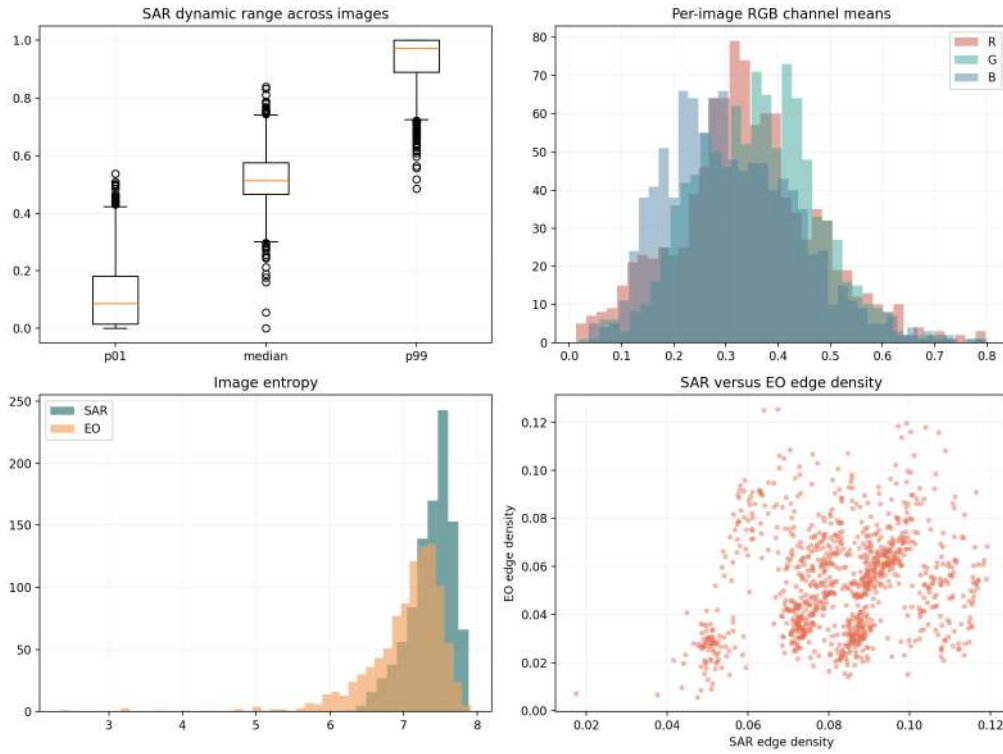
Figure 3: Dataset composition. The terrain classes are balanced, and the split is scene-disjoint to reduce leakage.

4.2 Image Statistics and Quality

The SAR display images and EO RGB images are normalized to $[-1, 1]$ before training. The EDA reports SAR mean 0.524, SAR standard deviation 0.179, EO brightness 0.336, SAR entropy 7.371, and EO entropy 7.029 over the sampled statistics set. Since the dataset uses PNG display imagery rather than calibrated Sentinel-1 backscatter, the project does not claim physically calibrated radiometry.



(a) SAR and RGB pixel histograms.



(b) Image-level statistics.

Figure 4: EDA statistics used to justify normalization and expose modality differences.

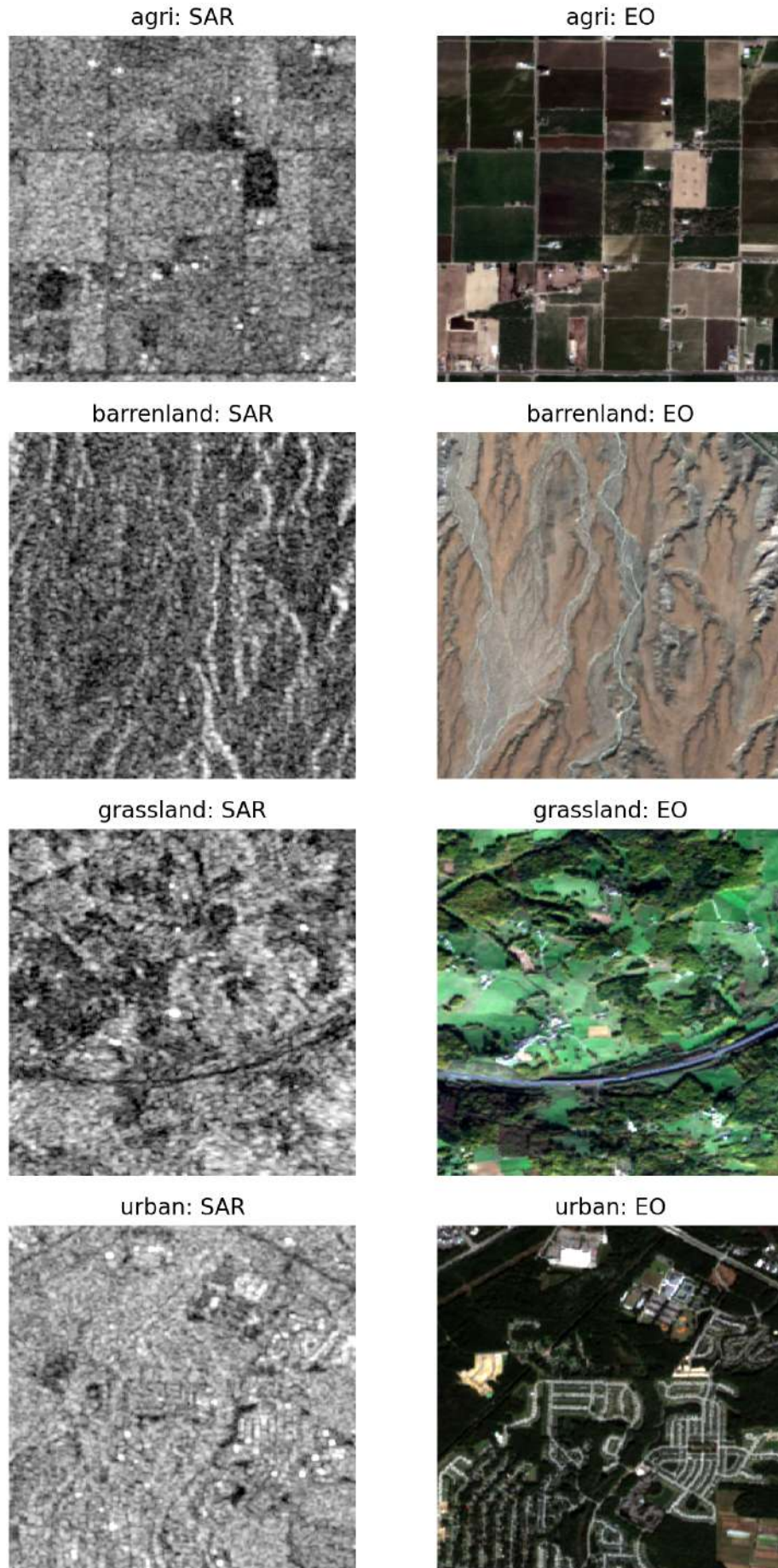


Figure 5: Terrain-wise SAR and EO examples. These examples show why SAR-to-EO is underdetermined: SAR structure can be visible, but optical colour and material texture are not uniquely specified.

5 Experimental Design

5.1 Progressive Model Sequence

The five models are intentionally ordered as a research progression. Each stage introduces one major modelling idea and tests a specific hypothesis.

Table 2: Progressive model design.

Stage	Architecture	Objective	Research question
M1	U-Net	L1	How far does deterministic paired reconstruction go?
M2	U-Net + PatchGAN	BCE GAN + L1	Does adversarial learning improve EO realism?
M3	Pix2Pix + perceptual/FM	BCE GAN + L1 + perceptual + FM	Do feature losses reduce over-smoothing?
M4	Attention ResUNet + spectral PatchGAN	Hinge GAN + L1 + perceptual + FM	Does attention improve structure?
M5	Attention ResUNet + spectral PatchGAN + EMA	Hinge GAN + L1 + perceptual + FM + gradient	Can a final paired model improve robustness?

5.2 Evaluation Protocol

All models are evaluated with the same manifest, image size, preprocessing, and held-out splits. Metrics are reported separately on validation and test subsets. The common evaluation uses 200 samples per split, batch size 1, LPIPS enabled, and FID enabled. This subset size is a compute compromise: it is large enough to compare models consistently, but not a complete full-test benchmark. The full test split contains 2,711 samples and should be used for a final production-grade benchmark when sufficient GPU time is available.

The reported metrics are:

- PSNR: pixel-level reconstruction fidelity; higher is better.
- SSIM: structural similarity; higher is better.
- MAE: absolute pixel error; lower is better.
- LPIPS: learned perceptual distance; lower is better.
- FID: distributional distance between generated and real EO images; lower is better.
- Runtime: milliseconds per image on the local evaluation environment; lower is better.

6 Training and Resource Log

Table 3: Training summary from the completed 30-epoch runs. Lower validation L1 is better.

Model	Epochs	Best epoch	Best val. L1	Final val. L1	Time
M1 U-Net L1	30	11	0.3882	0.4555	47.7 min
M2 Pix2Pix	30	9	0.4360	0.4662	79.9 min
M3 Pix2Pix + perceptual/FM	30	24	0.4002	0.4517	135.1 min
M4 Attention ResUNet	30	4	0.4067	0.4322	144.7 min
M5 Final Fusion GAN	30	8	0.4152	0.4392	147.1 min

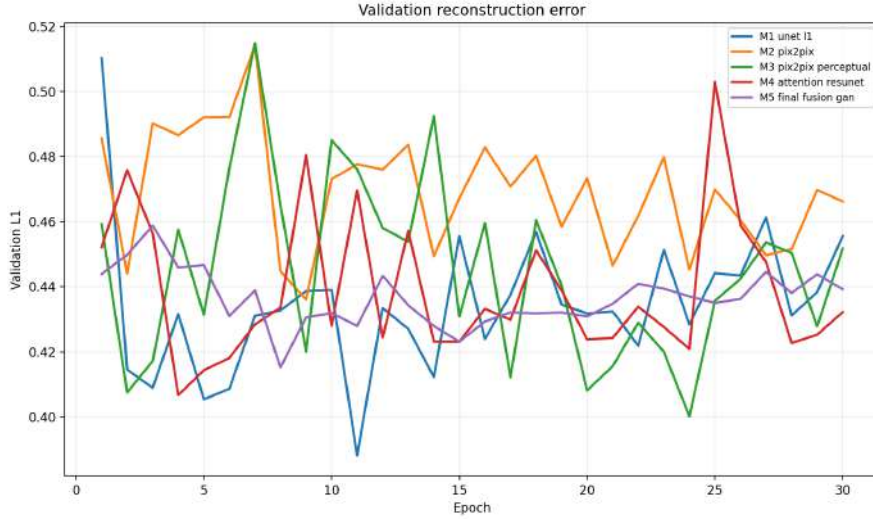


Figure 6: Training curves generated by the report script. M1 optimizes the L1 objective most directly, while later GAN variants trade some reconstruction fidelity for perceptual and distributional behaviour.

Training was performed under constrained notebook and local-compute conditions. Kaggle working storage resets required checkpoint archiving and local restoration of outputs. This affected the experimental strategy: rather than training one very large model, the project used a staged sequence of models and evaluated each with the same metric pipeline.

7 Validation and Test Results with FID

7.1 Validation Split Results

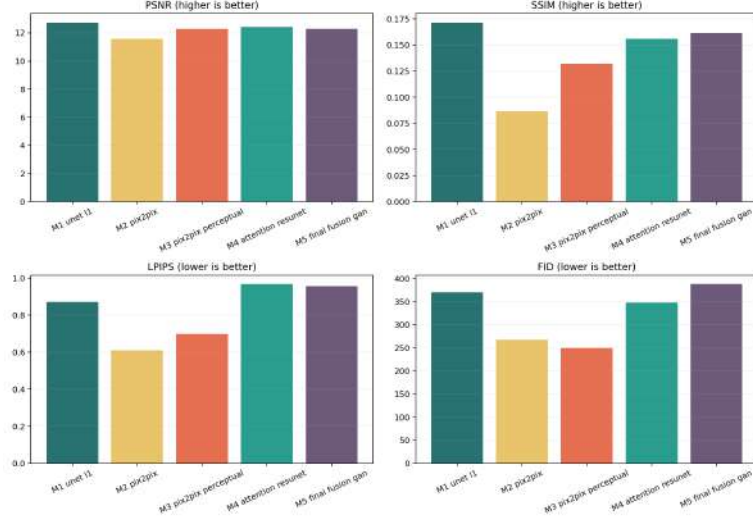
Table 4: Validation split evaluation with FID enabled. Metrics are computed on 200 held-out validation samples.

Model	Params (M)	PSNR $\uparrow$	SSIM $\uparrow$	MAE $\downarrow$	LPIPS $\downarrow$	FID $\downarrow$	ms/img $\downarrow$
M1: U-Net L1	54.41	12.730	0.171	0.203	0.872	369.888	214.84
M2: Pix2Pix	54.41	11.585	0.087	0.225	0.609	268.153	306.17
M3: Pix2Pix + perceptual/FM	54.41	12.282	0.132	0.212	0.698	249.734	248.60
M4: Attention ResUNet	51.34	12.421	0.156	0.213	0.967	348.377	255.17
M5: Final Fusion GAN	51.34	12.273	0.162	0.218	0.956	387.833	252.78

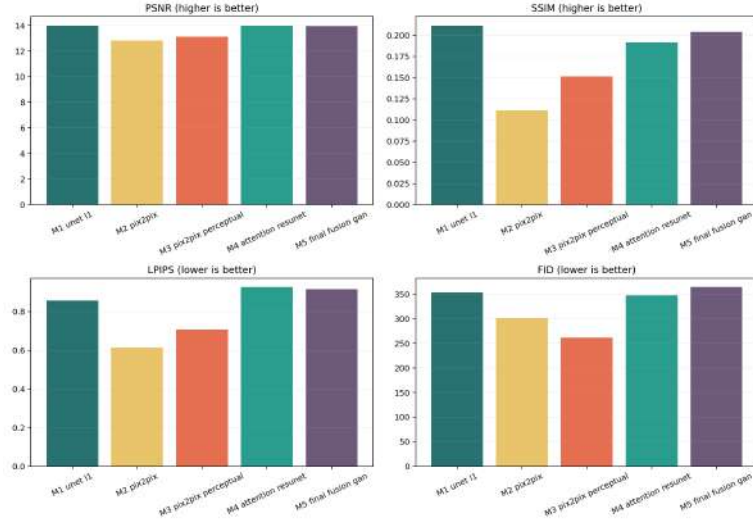
7.2 Test Split Results

Table 5: Test split evaluation with FID enabled. Metrics are computed on 200 held-out test samples.

Model	Params (M)	PSNR $\uparrow$	SSIM $\uparrow$	MAE $\downarrow$	LPIPS $\downarrow$	FID $\downarrow$	ms/img $\downarrow$
M1: U-Net L1	54.41	13.987	0.211	0.165	0.858	353.565	212.39
M2: Pix2Pix	54.41	12.833	0.111	0.183	0.615	301.522	247.33
M3: Pix2Pix + perceptual/FM	54.41	13.149	0.152	0.182	0.709	261.647	296.00
M4: Attention ResUNet	51.34	13.994	0.191	0.166	0.927	347.674	265.83
M5: Final Fusion GAN	51.34	13.963	0.204	0.164	0.917	364.020	267.47



(a) Validation metrics with FID.



(b) Test metrics with FID.

Figure 7: Validation and test comparison across PSNR, SSIM, MAE, LPIPS, FID, and runtime.

The validation and test results show that no single model dominates every criterion. On the validation split, the deterministic U-Net baseline achieves the best PSNR, SSIM, and MAE, indicating that direct supervised regression remains strong when the objective is low pixel error. However, Pix2Pix obtains the best LPIPS, while the perceptual/feature-matching model achieves the best FID. This suggests that adversarial and feature-space objectives improve distribution-level realism even when they reduce strict pixel alignment.

The same pattern appears on the test split. M4 gives the best PSNR, M1 gives the best SSIM, M5 gives the lowest MAE, M2 gives the best LPIPS, and M3 gives the best FID. Therefore, the project does not support the claim that the largest or most complex architecture is automatically best. Instead, the evidence supports a more careful conclusion: SAR-to-EO generation requires balancing structural preservation, optical realism, and stability. The best model depends on whether the downstream goal prioritizes geometry, perceptual plausibility, or distributional similarity.

7.3 Technical Summary of Results

- The dataset contains 16,000 paired SAR/EO patches from four terrain classes, with a scene-disjoint split to reduce leakage.
- The final evaluation uses PSNR, SSIM, MAE, LPIPS, FID, and runtime on 200-sample validation and test subsets.
- M1 is the strongest conservative reconstruction baseline, winning validation PSNR, validation SSIM, validation MAE, and test SSIM.
- M2 wins LPIPS on both validation and test, showing that adversarial learning improves feature-space perceptual similarity but weakens paired geometry.
- M3 wins FID on both validation and test, showing that perceptual and feature-matching losses improve distribution-level optical realism.
- M4 wins test PSNR but has weak LPIPS, showing that structural/pixel fidelity and perceptual realism do not necessarily agree.
- M5 wins test MAE and is the best final GAN compromise, but qualitative inspection shows remaining smoothing and colour bias.
- The controlled ablation shows that removing perceptual loss causes the clearest degradation, while attention is not automatically beneficial under short training.

7.4 Model-by-model Interpretation

7.4.1 M1 U-Net L1

M1 is a deterministic U-Net trained only with L1 loss. It is the strongest conservative baseline: it wins validation PSNR, validation SSIM, validation MAE, and test SSIM. It is also the fastest model in local test evaluation. Qualitatively, its outputs are stable but smooth. This confirms the expected behaviour of L1 training: the model learns average optical appearance and preserves broad layout, but loses high-frequency texture and target-specific colour.

7.4.2 M2 Pix2Pix

M2 adds a PatchGAN discriminator to the U-Net generator. It achieves the best LPIPS on both validation and test, showing that adversarial learning improves feature-space perceptual similarity. However, it has weaker PSNR and SSIM. Visual outputs are sharper than M1 but often contain texture that is not aligned with the EO target. This is the classic Pix2Pix trade-off: local realism can improve while paired geometry degrades.

7.4.3 M3 Pix2Pix + Perceptual/FM

M3 adds perceptual and feature-matching losses. It achieves the best FID on both validation and test, making it the strongest distribution-level model. It improves over M2 in PSNR and SSIM while keeping better perceptual behaviour than the deterministic baseline. This result supports the idea that feature-space supervision is useful for SAR-to-EO generation, especially when evaluating the generated EO distribution rather than only pixel error.

7.4.4 M4 Attention ResUNet

M4 introduces an attention ResUNet generator and spectral-normalized PatchGAN discriminator. It achieves the best test PSNR and competitive MAE. However, it has weak LPIPS. This suggests that attention and residual structure improve average reconstruction, but do not guarantee perceptual realism. Qualitatively, M4 is coherent but often overly smooth and green-biased.

7.4.5 M5 Final Fusion GAN

M5 adds gradient consistency and EMA checkpointing to the attention GAN. It achieves the best test MAE and is the final GAN candidate. However, it does not dominate FID or LPIPS. The final model is therefore best described as the strongest average-error compromise, not as a complete perceptual solution. Its value is that it represents an evidence-driven endpoint after testing simpler baselines, adversarial learning, feature losses, attention, and structural regularization.

8 Controlled Ablation Study

To satisfy the controlled ablation requirement, reduced-budget variants of the final architecture were trained under a fixed setup. Each ablation removes one component from the final fusion model while keeping the remaining data split, training limit, evaluation limit, and model family fixed. The saved ablation artifact contains finite PSNR, SSIM, MAE, parameter count, and runtime values; LPIPS and FID were not finite in the ablation CSV and are therefore not reported for this table.

Table 6: Controlled ablation study on the final architecture under a reduced training budget. Metrics are computed on 200 test samples.

Variant	Params (M)	PSNR $\uparrow$	SSIM $\uparrow$	MAE $\downarrow$	ms/img $\downarrow$
Full final model	51.34	13.819	0.186	0.173	10.10
No attention	51.27	13.942	0.194	0.170	7.92
No perceptual loss	51.34	13.585	0.164	0.175	8.01
No feature matching	51.34	13.812	0.188	0.173	8.01
No gradient consistency	51.34	13.827	0.178	0.172	8.00

The ablation reveals three useful behaviours. First, removing perceptual loss causes the clearest degradation in PSNR and SSIM, suggesting that feature-space supervision helps stabilize the generator even in short training. Second, removing feature matching produces only a small change in this reduced run, which suggests that its stabilizing effect may need longer adversarial training to become more visible. Third, the no-attention variant performs best in the short ablation setting. This does not prove attention is harmful; it shows that attention increases complexity and may not help unless the training budget is sufficient for the model to learn how to use it.

Overall, the ablation supports the final design logic but also shows that every added component has a cost. Perceptual supervision is useful, feature matching is a stabilizer, gradient consistency trades pixel intensity for structural regularity, and attention should be treated as conditional rather than automatically beneficial.

9 Qualitative and Terrain-wise Analysis

9.1 Qualitative Comparison

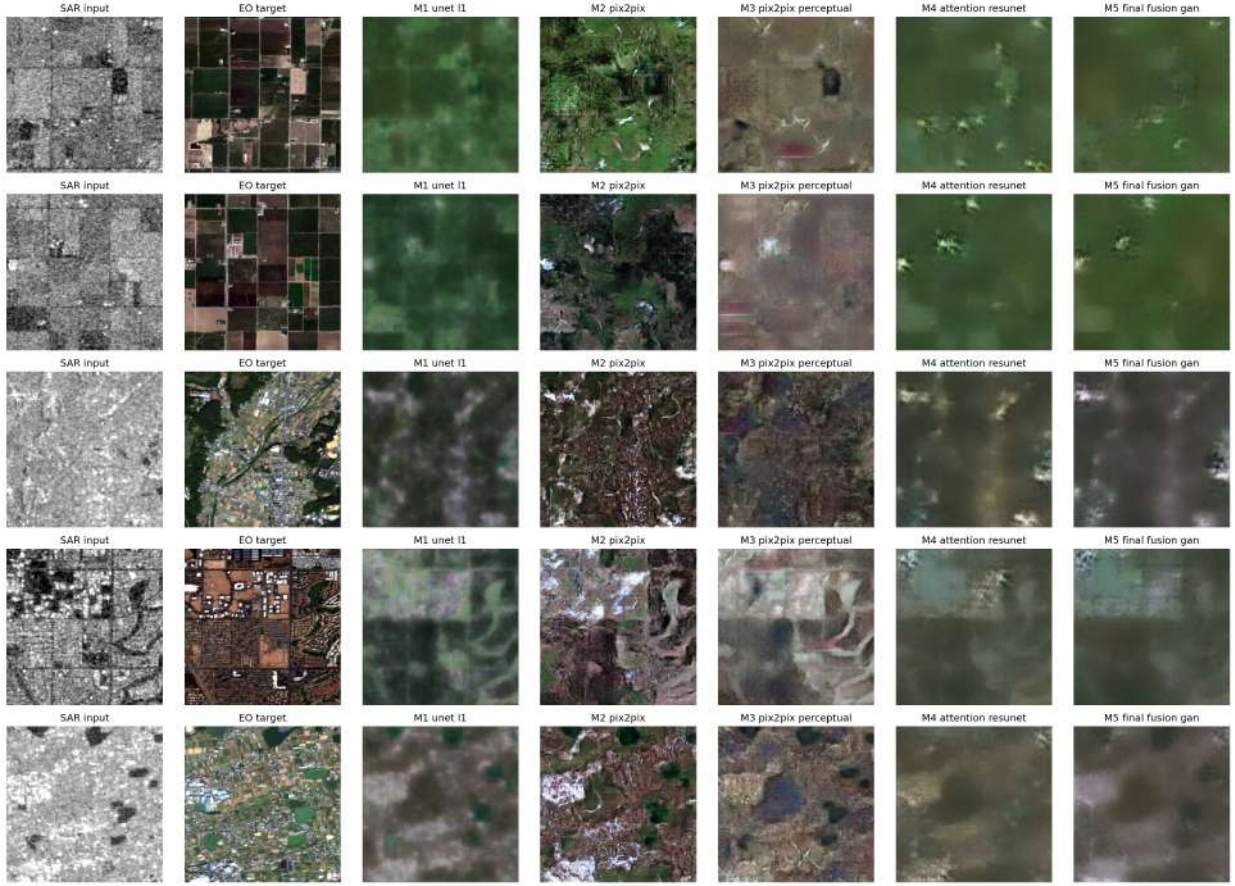


Figure 8: Qualitative comparison from the FID-enabled test evaluation run.

The qualitative grid explains the metric disagreement. M2 often looks more textured and therefore receives the best LPIPS, but its generated textures are not reliably aligned with the target. M1 and M5 are smoother and therefore less visually striking, yet they better control average error. M3 best matches the target EO distribution under FID. M4 gives high PSNR but weaker perceptual distance. This confirms that SAR-to-EO synthesis needs both fidelity and uncertainty-aware realism, not only a stronger GAN loss.

9.2 Terrain-wise Generalisation

Table 7: Terrain-wise held-out test results from the common evaluation.

Model	Terrain	Count	PSNR	SSIM	MAE	LPIPS
M1 U-Net L1	agri	64	14.941	0.3238	0.1451	0.8881
M1 U-Net L1	barrenland	14	11.180	0.1406	0.2395	0.7946
M1 U-Net L1	grassland	32	15.363	0.2852	0.1519	0.7802
M1 U-Net L1	urban	90	13.256	0.1164	0.1712	0.8739
M2 Pix2Pix	agri	64	13.683	0.1515	0.1629	0.6547
M2 Pix2Pix	barrenland	14	10.281	0.0955	0.2616	0.6411

Model	Terrain	Count	PSNR	SSIM	MAE	LPIPS
M2 Pix2Pix	grassland	32	13.817	0.1534	0.1654	0.5976
M2 Pix2Pix	urban	90	12.276	0.0703	0.1902	0.5892
M3 Pix2Pix + perceptual/FM	agri	64	13.348	0.2115	0.1815	0.7471
M3 Pix2Pix + perceptual/FM	barrenland	14	11.529	0.1261	0.2287	0.6340
M3 Pix2Pix + perceptual/FM	grassland	32	13.400	0.1981	0.1842	0.6126
M3 Pix2Pix + perceptual/FM	urban	90	13.170	0.0965	0.1735	0.7272
M4 Attention ResUNet	agri	64	14.612	0.2800	0.1530	0.9479
M4 Attention ResUNet	barrenland	14	11.033	0.1421	0.2470	0.8921
M4 Attention ResUNet	grassland	32	15.064	0.2554	0.1568	0.8778
M4 Attention ResUNet	urban	90	13.634	0.1135	0.1651	0.9346
M5 Final Fusion GAN	agri	64	14.261	0.2905	0.1579	0.9265
M5 Final Fusion GAN	barrenland	14	10.988	0.1482	0.2508	0.8851
M5 Final Fusion GAN	grassland	32	15.453	0.2904	0.1480	0.8536
M5 Final Fusion GAN	urban	90	13.684	0.1206	0.1612	0.9369

Terrain analysis gives a more nuanced result than the global table. M1 is strongest on agriculture under several pixel metrics. M3 is strongest on barren land under PSNR, MAE, and LPIPS. M5 is strongest on grassland and urban under PSNR, SSIM, and MAE. This suggests that the final model is not globally dominant, but it improves robustness on terrain groups containing more spatial variability and urban structure.

10 Failure Analysis

Script 09 shows that all models have the same lowest-SSIM terrain: urban. The strongest failure association for every model is the negative relation between EO edge density and SSIM. High-frequency target content such as building edges, roads, field boundaries, and sharp terrain details is systematically difficult.

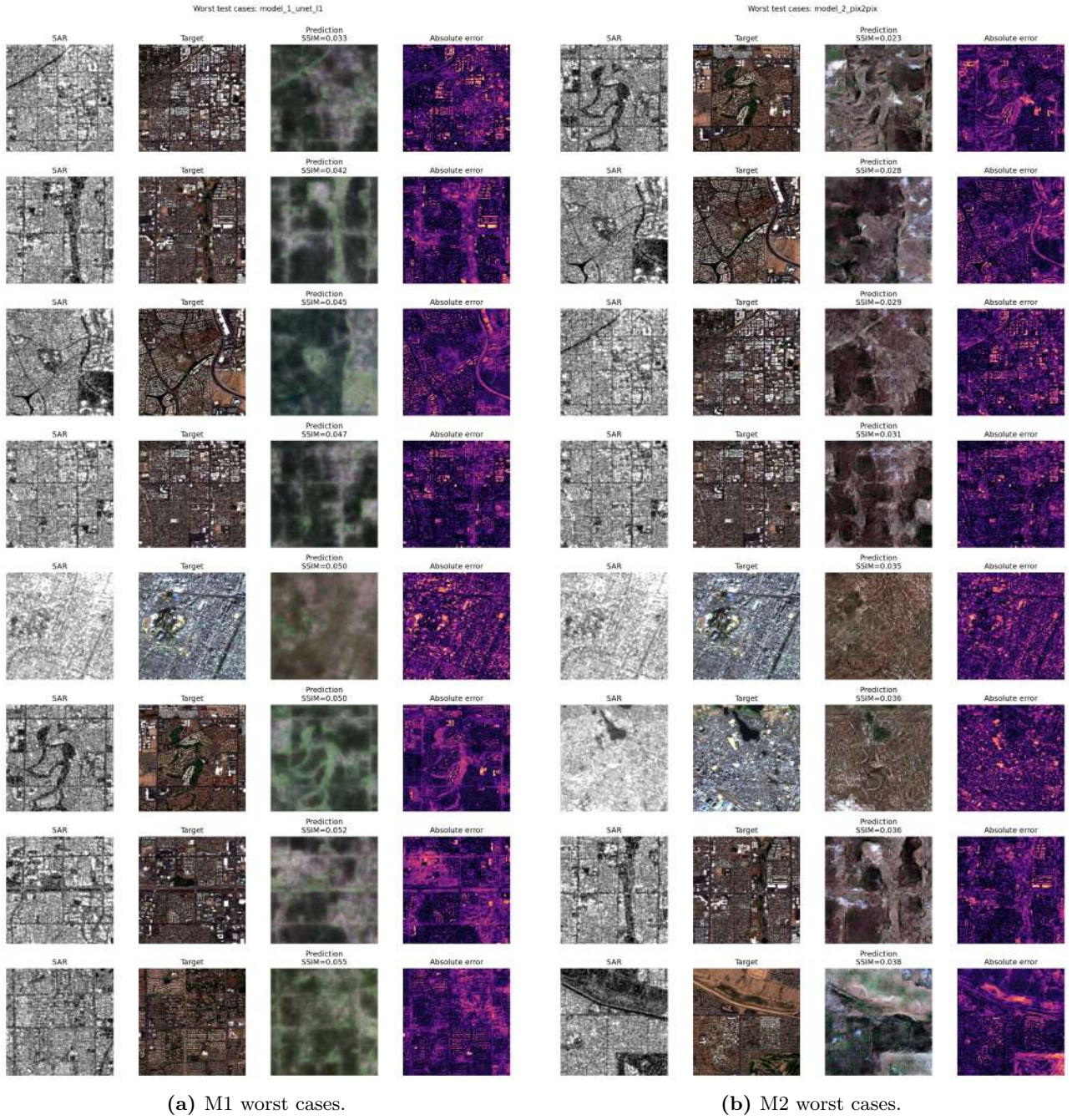


Figure 9: Worst-case analysis for M1 and M2. M1 fails by smoothing structure; M2 fails by producing sharper but less target-aligned texture.

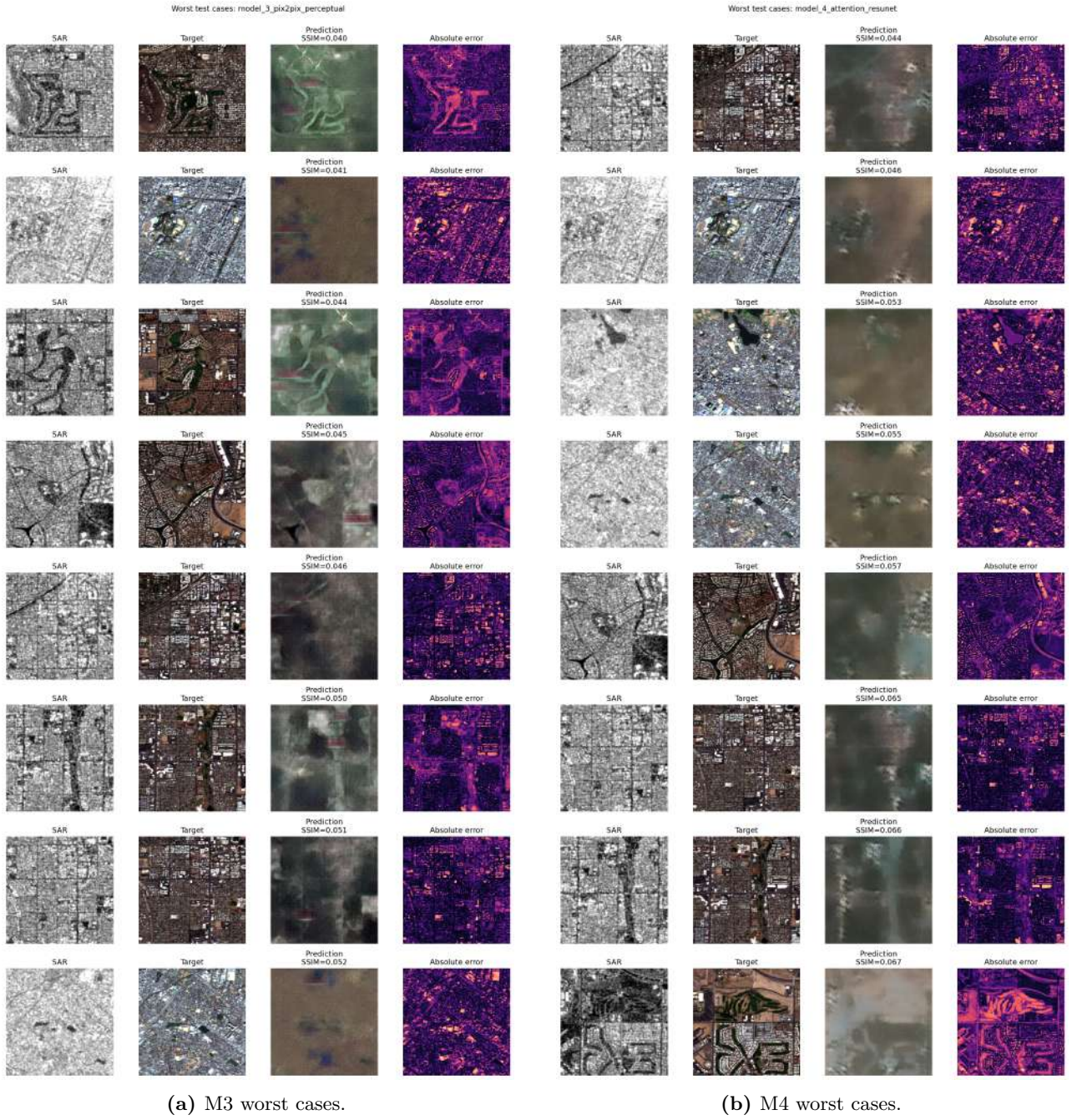


Figure 10: Worst-case analysis for M3 and M4. Feature losses and attention improve stability but do not recover dense EO structure.

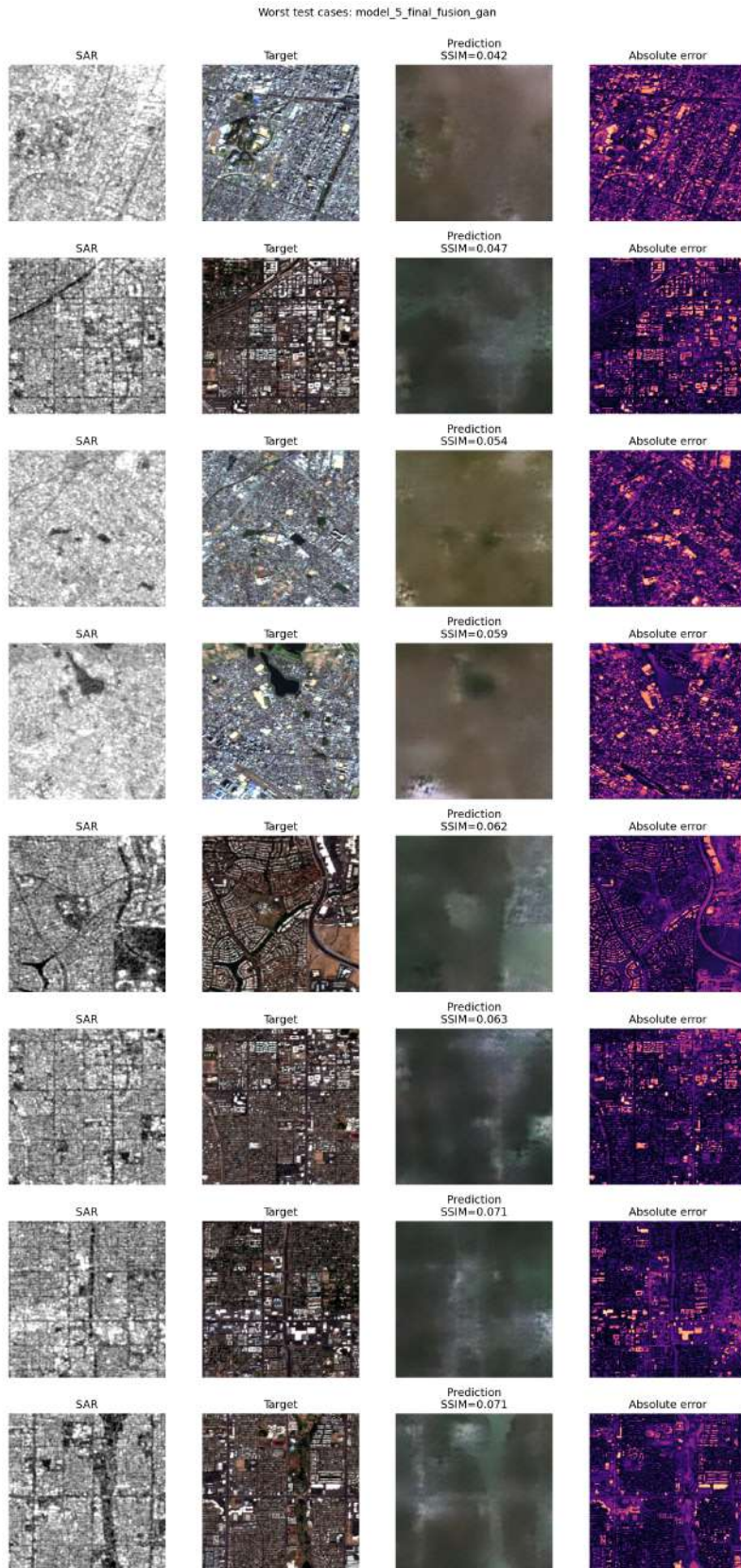


Figure 11: Worst-case analysis for M5. The final model reduces average error but still misses high-frequency urban and boundary detail.

The failures are not only implementation weaknesses; they follow from sensing physics and objective functions. First, SAR backscatter does not uniquely determine optical RGB colour. Second, L1 and MAE objectives reward central tendency and therefore encourage smooth outputs. Third, adversarial

objectives reward local realism but can hallucinate target-inconsistent texture. Fourth, the dataset uses PNG display images rather than calibrated SAR backscatter, limiting physically meaningful preprocessing. Fifth, dense urban scenes contain many edges and small objects, exactly the content most difficult to infer from a single SAR patch.

11 Discussion

11.1 What Was Learned from Each Stage

The progressive design produced a clear learning path. M1 established that simple paired supervision is hard to beat under pixel and structure metrics. M2 showed that adversarial learning improves LPIPS but can damage geometry. M3 showed that perceptual and feature-matching losses improve FID and distributional realism. M4 showed that attention and spectral normalization improve PSNR and coherence but do not guarantee perceptual similarity. M5 showed that gradient consistency and EMA improve average test error, but do not fully recover target-specific optical detail.

11.2 Best Model Selection

If the selection criterion is validation stability and structural reconstruction, M1 is the best model. If the criterion is test PSNR, M4 is best. If the criterion is test MAE and final GAN robustness, M5 is best. If the criterion is LPIPS, M2 is best. If the criterion is FID, M3 is best. Therefore, the recommended final solution is not framed as “one model solves SAR-to-EO”. The defensible result is a model family comparison where M5 is the final evidence-driven GAN candidate and M1 remains the strongest conservative reconstruction baseline.

11.3 Connection to the Literature

The observed behaviour is consistent with the original image-translation literature. Pix2Pix-style models can improve visual texture but may reduce paired fidelity. Perceptual and feature-matching losses improve feature-level realism and distributional behaviour. Attention and residual modules can improve reconstruction but add complexity. Diffusion models are a logical next step because they can represent multiple plausible EO outputs for one SAR input, which better matches the uncertainty of the problem.

12 Limitations

The main limitation is the underdetermined one-to-many nature of SAR-to-EO translation. A generated EO patch may be plausible but still wrong. The second limitation is data representation: the dataset contains PNG display imagery, not calibrated SAR backscatter. The third limitation is computational: FID was computed on 200-sample validation and test subsets, not on the full 2,711-sample test split. The fourth limitation is that the ablation was run under a reduced budget, so it is useful for isolating short-run component effects but should not be interpreted as a full-budget architecture search. Finally, LPIPS and perceptual losses rely on natural-image feature spaces, which are not perfect for remote sensing imagery.

13 Future Work

The immediate next step is to run the full 2,711-sample test evaluation with FID on a stronger GPU. The second step is full-budget ablation training for the final model, including finite LPIPS and FID for each ablation. The third step is to use calibrated Sentinel-1 VV/VH backscatter rather than PNG display intensities. The fourth step is to add terrain conditioning, structure-aware losses, segmentation priors, or edge-preserving objectives. The final research direction is conditional or latent diffusion. Diffusion is attractive because it can sample multiple plausible optical outputs for a single SAR input, which is more faithful to SAR-to-EO uncertainty than a single deterministic RGB prediction.

14 Conclusion

This project builds a research-quality SAR-to-EO translation workflow. It analyzes the dataset, constructs a reproducible paired pipeline, trains five progressively motivated models, evaluates them on validation and test subsets with FID, and performs controlled ablation and failure analysis. The final results are honest and technically useful: no model dominates all metrics. M1 is best for conservative structural reconstruction, M2 is best for LPIPS, M3 is best for FID, M4 is best for test PSNR, and M5 is best for test MAE. The strongest conclusion is that SAR-to-EO translation requires explicit treatment of uncertainty, structure preservation, and modality ambiguity. The work therefore provides a solid baseline and a defensible path toward calibrated, diffusion-based, or uncertainty-aware generative models.

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